Introduction to machine learning
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Genetic algorithm: learn from nature

- recombination
- mutation
- selection of fittest (competition)

Representation of the solution / behaviour by Chromosomes. (e.g. weights of a NN)

The fitness of the chromosome can be evaluated.

eg: games: chromosomes = strategy
fitness = score

The fitness is often more than just a score, we can reward other behaviour as well
eg: chess: material advantage, position

Chromosomes are made of genes
whole genes can be exchanged between chromosomes, and the genetic code remains valid, crossover must not destroy genetic code.

Genes can be of variable length so can chromosomes

Ex: Travelling salesman: city order → bad

Ordinal:	1 2 3 4	3	3
	1 2 4	3	3 4
	1 2	2	3 4 2
	1	1	3 4 2 1

Once the fitnesses are calculated, the crossover procedure starts

A fraction (often 1/2) of the population is selected for reproduction, the rest is killed

Selection mechanism

- Rank selection: rank according to fitness and keep top $\frac{1}{2} \cdot N$
  + best is always kept
  + may converge fast
  - easily trapped in a local minimum
- Roulette wheel selection: select with probability prop. to f
  + best are likely kept
  + can get out of local minima
  - bad for the late stage if f is being changing
  - sensitive to f scale
- linear rank selection:

  $sp \in [1, 2]$ $\quad r_i$: rank of cho. i.

  $P(r_i) = \frac{1}{N}\left(sp - (2sp-2)\frac{i-1}{N-1}\right)$

  $sp = 1 \rightarrow$ random

  $sp = 2 \rightarrow$ linear rank selection

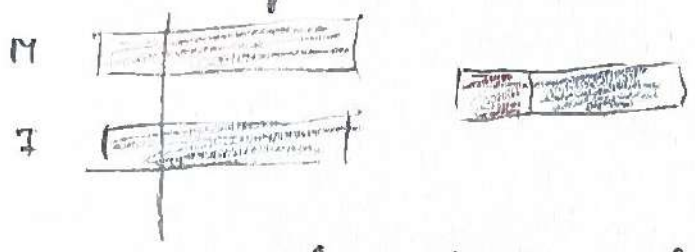
  + tunable
  + can get out of local minima

From the selected chromosomes pairs are selected to create an offspring

Crossover

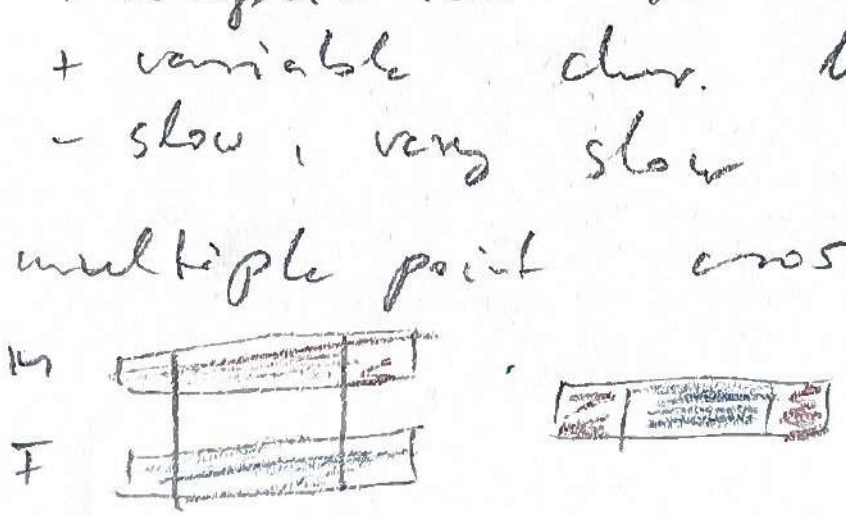
Two parents → one offspring

- single point crossover



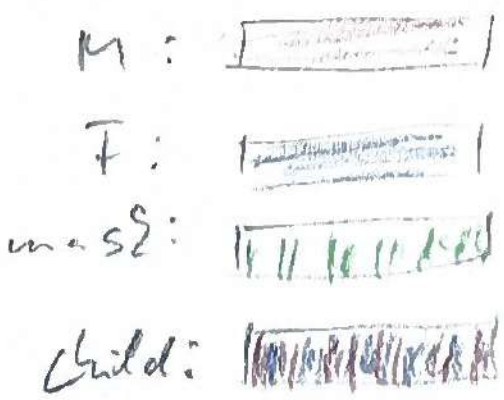
  + complicated gene structure
  + variable chr. length
  - slow, very slow

- multiple point crossover



  ↓ convergence can be faster
  - algorithm gets slower

- random crossover



  + fast convergence
  + fast algorithm
  - variable chr. length

Mutation: change gene randomly with p_m

Implementation:

Get n_m from binomial distribution
with N, p_m

Select n_m random genes in general
if n_m is small we do not care
about replacement

change selected genes